

Danny McCulloch

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University of Exeter | PhD researcher | Department of Mathematics & Statistics

KEY POINTS AND ACHIEVEMENTS

- Awarded 3 out of 4 highly competitive travel grants (combined value of £2.6k)
- Published paper in 'Geoscientific Model Development' (Impact factor 2022: 5.1) and preprint in Journal of Advances in Modeling Earth Systems (Impact factor 2025: 4.6)
- Awarded the 'Deans commendation for academic citizenship' for a tutorial series I developed and delivered for undergraduates
- Co-organiser for the Cloud Feedback Model Intercomparison Project (CFMIP) meeting, 2025
- Presented at 12 research conferences and 5 NERC-funded project workshops
- Extensive (5+ years) teaching experience across several programs
- Lead for professional development coordination for the Exeter university Exoplanet Theory group
- Fresh eyes on CMIP subgroup co-lead

FUNDING AWARDS

University of Exeter Physics and Astronomy and Natural Sciences Travel Bursary <i>Competitive travel bursary to cover conference costs</i>	March 2024 £500
Fresh Eyes on CMIP (FeCMIP) travel award <i>Competitive travel award open to all FeCMIP members across 17+ countries</i>	March 2024 ~£1600
Professor Ozgur Akman travel bursary <i>Competitive travel bursary to cover conference costs</i>	February 2024 £500
PhD studentship <i>Highly competitive PhD position funded for 3.5 years</i>	September 2022 ~£70,000

EDUCATION

PhD in Natural Sciences Supervisors: Prof. Hugo Lambert, Dr. Mark Webb, Prof. Geoffrey Vallis	Sep 2022 – Jun 2026 University of Exeter, UK
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Thesis title: "*Understanding and constraining the relationship between cloud feedbacks, the tropical circulation and the hydrological cycle*"

Research: Testing the subtropical cloud response to tropical atmospheric perturbations with the aim of improving circulation representation caused by climate change. Involves using and modifying the Unified Model (UM), a Global Climate Model (GCM) used extensively on Earth climate modelling and in CMIP. This involved:

- Model development and source code modification to induce targeted prognostic changes within the dynamical simulations via the radiative transfer scheme
- Analysis and creating interactive projections of GCM simulations using Python
- Effective and efficient self-management – coordinating and planning my own research goals/tasks, as well as seeking new project and funding opportunities, throughout the PhD
- Presentations at both national, and international conferences relevant to my field about my results.
- Integration in groups that are focused on *how* models can be applied outside of research
- Supervising MSc students who are furthering my prior work on a Martian climate model

- Teaching experience (5+ years) across a variety of scientific disciplines, ranging from primary schools to 2nd year undergraduates

MSc by Research in Physics

Supervisors: Prof. Nathan Mayne, Prof. Matthew Bate, Dr Denis Sergeev

Sep 2020 – Sep 2022

University of Exeter, UK

Thesis title: *"Modelling an idealised modern-Mars climate using the Met Office Unified Model"*

Research: Adapted and tuned the UM to simulate a modern-day Mars climate. This involved:

- Adapted the UM to characterise Mars' current climate, which involved taking several climate processes and altering/tuning them in the UM. This required extensive knowledge on how models work and what is needed/approximated/not needed to accurately simulate a planets climate
- Publication for the journal *"Geoscientific Model Development"*
- Detailed verification and comparison of the adapted UM output to other GCM outputs and in-situ data
- Hypothetical terraforming scenarios using the newly-adapted Mars UM and investigated the subsequent effects

BSc Zoology

2:1 with Honours | Dr Bob Brewin, Dr Alejandro Roman Gonzalez

Sep 2017 – Jul 2020

University of Exeter, UK

Dissertation title: *"Does a decrease in sea ice lead to an increase of phytoplankton in the Southern Ocean?"*

Research:

- Analysis and manipulation of Antarctic sea ice & chlorophyll-a satellite data (predominantly from the AQUA-MODIS instrumentation)
- Statistical analysis on temporal and spatial changes to both aforementioned parameters individually and with their relationship to each other, with a focus on investigating how climate change might affect these parameters in the future
- Demonstrated key ocean circulation and fluid dynamics processes and wrote up problem sheets for how these processes work, which led to an internship (e.g. Ekman transport, Taylor-Proudman theorem and solid body rotation)

PUBLICATIONS AND PRESENTATIONS

All viewable at [ORCID ID: 0000-0002-6260-9570](https://orcid.org/0000-0002-6260-9570)

Preprint – Disentangling cloud/circulation interactions (PhD): McCulloch, D., *et al* (2025): *"Weakening Tropical Deep Convection Reduces Subtropical Low Clouds via Lower Free-Tropospheric Moisture Convergence in a Climate Model"*, Authorea., DOI: 10.22541/au.175407740.01003894/v1

Paper – First application of the UM to Mars (MbyRes): McCulloch, D., *et al* (2023): *"A modern-day Mars climate in the Met Office Unified Model: dry simulations"*, *Geoscientific Model Development*., DOI:10.5194/gmd-16-621-2023

Micro-publication from the group I taught and supervised: R J, A. S., Marvi, K., Phatnani, K. S., Sordilla, N. M., Maure, E., & McCulloch, D. (2024): *"Impacts of ENSO on Climate Variables in the Pacific Region"*, *Climatematic Impact Scholars Program 2023-2024 (CISP)*., DOI:10.5281/zenodo.11201102

Reviewer for *Geophysical research letters*, 2024

Presentations: _____

CFMIP: Paris 2023, Boston 2024 and **Exeter 2025** (co-organiser)

European Geophysics Union (EGU) general assembly: May-2022, April-2024, April-2025 (DOIs on ORCID)

Invited speaker – Met Office Seminar Series, Dec-2024, Dec-2023 and July-2022

Mars Atmosphere Modelling and Observations workshop, June-2022: Mcculloch, D. et al. (2022) *"Simulating Mars' Climate using the Met Office Unified Model"*, in the Seventh International Workshop on the Mars Atmosphere: Modelling and Observations, p. 1502.

UK Exoplanet Meeting (UKEXOM): April-2021

Euro Planet Science Congress (EPSC): September-2021 ([DOI](#))

ADDITIONAL EXPERIENCE AND OUTREACH

Fresh eyes on CMIP

Improving CMIP resources to improve accessibility and approachability

Sep 2023 – current
Model evaluation subgroup co-lead

Deep Learning for Weather and Climate Applications

A Met Office 1-week course teaching ML techniques to climate scientists

May 2025 and Dec 2025
Teaching assistant

Exoplanet Theory Group (ETG) training lead

Organising and managing workshops on effective team development

July 2023 – July 2025
Professional development & social sec.

Climate match academy

2-week summer school teaching comprehensive climate science

July 2023, 2024 & 2025
Lead teaching assistant

"Frontiers in Science" – 2nd year Natural Sciences module

Involved designing and presenting guides on scientific communication, marking and module evaluation in the interest of content improvement.

Sep 2020 – present
Lead Postgraduate Teaching Assistant

Exeter Science Centre events

Various shows and events promoting climate science to Exeter locals

Mar 2022, Jun 2024, Apr 2024, Jul 2025
Roaming scientist

"Careers in Extreme Weather" (YouTube video) by the University of Bristol

Outreach information for A-level students in Devon about what a job in climate looks like

Sep 2024
Myself

Soundart Radio: "Everybody Counts: Using Climate Models"

Radio series to make mathematics more approachable and why its important in everyday life

Sep 2023
Invited speaker

"Science in the square" outreach event

Showing climate science and renewable energy concepts to Falmouth locals

Jun 2019
Demonstrator

Science experiment teacher

Outreach program visiting schools and teaching simple experiment techniques

Mar–Jun 2018–2020
Teacher

British army – Exeter University Officer Training Corps

Effective time management and work under stressful conditions, developing mental resilience

Sep 2017 – Jun 2020
Officer cadet

SKILLS

Languages: English (native), Spanish (fluent) – **Coding:** Python, FORTRAN, R, Unix systems

Technical: Numerical Global Climate Models development, Python proficiency [Iris, Xarray, tensorflow, SciPy, NumPy, Matplotlib, PyVista], version and subversion control [Git, SVN], large data analysis (>1Tb), extensive experience in organising and chairing scientific sessions

Personal hobbies: Electronics and robotics, Snowboarding, DIY projects, hydroponics, boardgames